
NSR-KT: Disentangling Sequential Regularities via Neuro-Symbolic Residual Learning in Knowledge Tracing

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Abstract

Knowledge tracing (KT) aims to predict a learner’s future performance from historical interactions. Most existing neural approaches model the final response probability directly, which can entangle simple sequential regularities with more complex effects arising from item identity and auxiliary information. We propose NSR-KT, a neuro-symbolic residual framework that explicitly separates these two sources of signal. A lightweight symbolic component provides an interpretable probabilistic prior capturing simple sequential regularities, while a neural backbone learns residual corrections in logit space to model systematic deviations beyond the symbolic structure, allowing it to focus on higher-order patterns without redundantly relearning simple behaviors. To further improve robustness, we extend the residual formulation to multiple complementary experts and combine them through a global convex aggregation scheme with weights selected on held-out folds, enabling a flexible expert combination mechanism. We provide a theoretical perspective on the prior-plus-residual formulation and evaluate NSR-KT on several public KT benchmarks, where it consistently improves predictive performance over strong neural baselines while offering a more structured view of how different modeling components contribute to final predictions.

1 Introduction

KT studies how to infer a learner’s evolving mastery state from a sequence of responses, question identities, and auxiliary behavioral signals. The field has progressed from Bayesian Knowledge Tracing (BKT) to deep recurrent models, self-attention models, and recent Transformer variants that substantially improve predictive accuracy on public educational benchmarks [Corbett and Anderson, 1994, Piech et al., 2015, Pandey and Karypis, 2019, Ghosh et al., 2020, Liu et al., 2023]. Still, two gaps remain.

First, most high-performing KT models are optimized as generic sequence predictors that directly output a final probability. They can fit subtle patterns, but the resulting representation often mixes simple local regularities with dense residual effects in a single black-box state. Second, strong neural backbones are data hungry. In cold-start or long-tail regimes, they may have too little evidence to form stable item and concept representations even when the learner history contains short repeated symbolic motifs.

The slogan “prediction is compression” is useful for this model-design problem. A learner trajectory contains recurring local patterns such as recovery after error streaks, stable success on recently mastered concepts, and concept-specific response motifs. If these regularities are predictable, they are also compressible. Symbolic sequence predictors such as PPM, PST, and context-mixing variants are designed to exploit repeated discrete contexts through explicit counts, suffixes, and expert mixtures. Their weakness is complementary to that of deep KT: they reason well about local symbolic history,

but they do not naturally absorb dense covariates, semantic item representations, or global transfer across related concepts.

This paper develops a neuro-symbolic alternative that treats KT as a prior-plus-residual problem. We propose NSR-KT, a two-module architecture in which a symbolic predictor supplies an explicit prior over the next response, while a neural KT backbone learns only the correction term needed to explain what the symbolic model misses. The symbolic branch is responsible for local, discrete, and interpretable regularities in learner histories; the neural branch is responsible for dense, global, and continuous structure that remains after symbolic compression. The framework is agnostic to the exact symbolic backend and is designed to accommodate PPM, PST, context mixing, and related finite-context predictors.

This residual interface also creates a natural extension beyond a single correction backbone. If several neural experts are trained above symbolic priors, their predictions can be combined with strictly fold-out-selected convex weights. This makes NSR-KT an ensemble of residual experts rather than a standalone neural ensemble: every expert is first residualized against a symbolic sequence prior, and the final combination exploits complementary residual corrections.

Our argument is not merely architectural. We also frame NSR-KT using the minimum description length (MDL) principle, which links predictive performance to the total code length required to describe the symbolic explanation, the residual correction, and the model itself. This perspective gives KT a useful inductive bias: a good predictor should let the symbolic module explain low-complexity sequential structure and let the neural module spend capacity on systematic residuals.

We make five main contributions:

1. We propose NSR-KT, a neuro-symbolic KT framework in which a compression-based symbolic predictor acts as a probabilistic prior and a neural network acts as a logit-space residual corrector.
2. We give a backend-agnostic interface that converts PPM, PST, context mixing, and related symbolic predictors into correctness priors usable by neural KT backbones.
3. We formalize why logit-residual fusion is a natural interface through an exact residual identity and an MDL-style interpretation.
4. We extend the interface to an ensemble of residual experts, where multiple neural residual predictors are combined with strictly fold-out-selected convex weights.
5. We evaluate NSR-KT on standard KT benchmarks using AUC and accuracy, including neural-ensemble baselines, symbolic-prior ablations, and residual-expert aggregation analyses.

2 Related Work

KT. Classical BKT models the acquisition of procedural knowledge with latent mastery transitions and observation noise [Corbett and Anderson, 1994]. Deep Knowledge Tracing (DKT) replaced manually specified state transitions with recurrent neural networks and established sequence modeling as the dominant KT paradigm [Piech et al., 2015]. Later models such as SAKT, AKT, and SimpleKT strengthened the neural side by improving long-range dependency modeling and interaction representations [Pandey and Karypis, 2019, Ghosh et al., 2020, Liu et al., 2023]. However, these methods still learn next-response probabilities in an end-to-end fashion, so repeated local response rules and high-level residual variation are absorbed into the same opaque predictor.

Symbolic sequence prediction and compression. Compression-based predictors convert next-symbol prediction into adaptive coding. PPM models a sequence through longest-match contexts with recursive escape and back-off [Cleary and Witten, 1984]. PST-style predictors retain a sparse suffix tree whose leaves summarize predictive contexts, while context-mixing variants combine multiple bounded-context experts rather than committing to a single suffix. These methods are attractive for KT because learner histories often contain short symbolic motifs such as repeated success after a concept switch or rapid recovery after a streak of errors. Yet such predictors are rarely used directly in KT because they do not naturally handle dense covariates, semantic item representations, or smooth transfer across concepts.

Neuro-symbolic and residual hybridization. Neuro-symbolic modeling aims to allocate complementary responsibilities to explicit symbolic structure and neural representation learning. In the KT setting, most prior hybrids inject handcrafted features, prerequisite graphs, or auxiliary relational

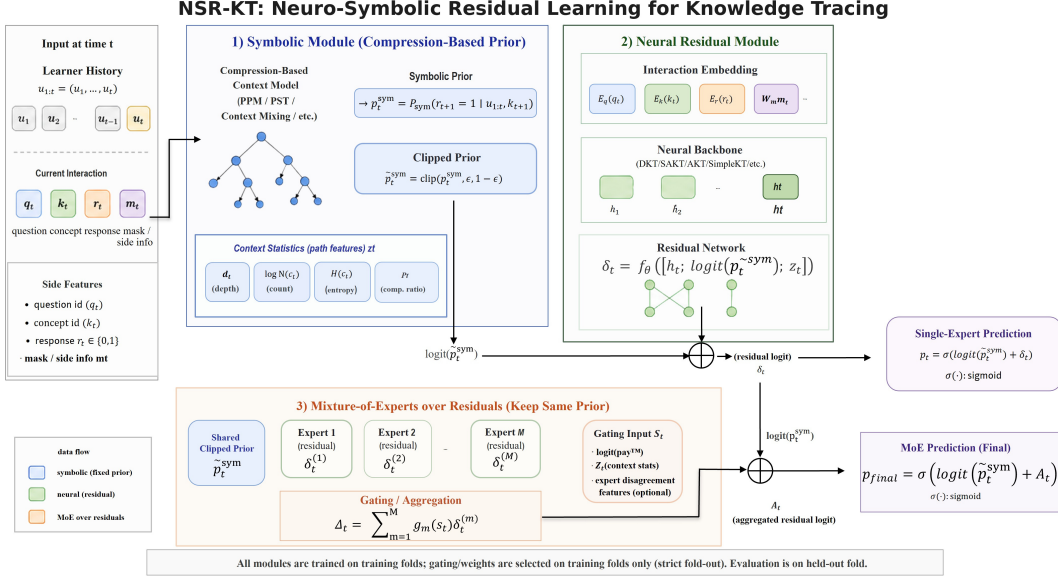


Figure 1: Overview of the NSR-KT framework. A symbolic finite-context predictor supplies a calibrated prior, and neural KT backbones learn residual logit corrections that can be combined by fold-out convex aggregation.

signals into a neural architecture, but they still let the neural model predict the final probability directly. Our formulation is more restrictive and therefore cleaner: the symbolic component is a standalone probabilistic predictor, and the neural component is asked only to fit the residual error of that predictor. This makes the interface between both modules explicit and analytically tractable. The ensemble variant keeps this structure at the expert level: each predictor is first residualized against a symbolic prior, and the final convex combination is selected under a strict fold-out protocol.

Minimum description length in learning. The MDL principle interprets learning as finding a concise description of both the data and the model used to encode it [Rissanen, 1978, Grünwald, 2007]. This lens is especially suitable for our setting because it clarifies why residual learning is desirable: if symbolic compression already captures the low-complexity portion of learner behavior, then the neural module should only pay description length for systematic mismatches that remain after compression.

3 Preliminaries and Theoretical Foundation

3.1 Preliminaries

Definition 1 (Learner trajectory). *For a learner with interaction horizon T , let*

$$\mathbf{x}_{1:T} = (x_1, x_2, \dots, x_T), \quad x_t = (q_t, k_t, r_t, m_t),$$

where q_t is the exercised item, k_t is the associated knowledge component, $r_t \in \{0, 1\}$ is the response correctness, and m_t denotes optional side information such as response time or attempt metadata.

Given the next queried item or concept (q_{t+1}, k_{t+1}) , the KT task is to estimate

$$p_t^* = P(r_{t+1} = 1 \mid \mathbf{x}_{1:t}, q_{t+1}, k_{t+1}).$$

Training typically minimizes the negative log-likelihood

$$\mathcal{L}_{\text{pred}} = - \sum_{t=1}^{T-1} \left[r_{t+1} \log p_t + (1 - r_{t+1}) \log(1 - p_t) \right]. \quad (1)$$

Symbolic tokenization. To expose explicit local response regularities, we map each interaction to a discrete token

$$u_t = \phi(q_t, k_t, r_t) \in \mathcal{A}.$$

A practical choice is to let the alphabet encode concept-response pairs, $\mathcal{A} = \mathcal{K} \times \{0, 1\}$, so that knowledge tracing becomes next-symbol prediction over concept-tagged outcomes.

Compression-based symbolic prediction. Let $c_t = u_{t-d:t-1}$ be the active suffix context with depth $d \leq D$. A generic symbolic predictor P_{sym} estimates probabilities over the next token u_{t+1} . When the next concept k_{t+1} is known, the symbolic prior over correctness is obtained by normalizing the two candidate tokens:

$$p_t^{\text{sym}} = \frac{P_{\text{sym}}(u_{t+1} = (k_{t+1}, 1) \mid u_{1:t})}{\sum_{r \in \{0, 1\}} P_{\text{sym}}(u_{t+1} = (k_{t+1}, r) \mid u_{1:t})}. \quad (2)$$

Representative instantiations include PPM [Cleary and Witten, 1984], PST [Ron et al., 1996], and context-mixing predictors [Mahoney, 2005]. NSR-KT only requires a causal normalized distribution over the next symbolic token, so the same residual interface can use different symbolic backends. Backend-specific choices affect only how P_{sym} is estimated; the residual interface itself is unchanged.

Residual parameterization. Our central modeling assumption is that the symbolic predictor should not be discarded. Instead, the final KT predictor is parameterized as a correction of the symbolic prior:

$$p_t = \sigma(\text{logit}(\tilde{p}_t^{\text{sym}}) + \delta_t), \quad (3)$$

where $\tilde{p}_t^{\text{sym}} = \text{clip}(p_t^{\text{sym}}, \epsilon, 1 - \epsilon)$ avoids degenerate logits and δ_t is a residual term learned by a neural network.

3.2 Theoretical Foundation

Let $\mathcal{X}$ denote the stochastic process that generates learner trajectories. Its entropy rate is

$$H(\mathcal{X}) = \lim_{T \rightarrow \infty} \frac{1}{T} H(X_{1:T}), \quad (4)$$

whenever the limit exists.

Proposition 1 (Logit-space residual decomposition). *For any target probability $p_t^* \in (0, 1)$ and any symbolic prior $p_t^{\text{sym}} \in (0, 1)$, there exists a unique residual*

$$\delta_t^* = \text{logit}(p_t^*) - \text{logit}(p_t^{\text{sym}}) \quad (5)$$

such that

$$p_t^* = \sigma(\text{logit}(p_t^{\text{sym}}) + \delta_t^*). \quad (6)$$

This identity gives the residual network a precise role: it does not need to relearn the entire predictor, only the mismatch between the symbolic prior and the unknown Bayes-optimal response probability.

MDL interpretation. The residual design has an MDL interpretation: the symbolic branch should encode low-complexity finite-context regularities, while the neural branch spends capacity only on systematic residual structure not captured by compression. This motivates regularizing both the symbolic context structure and neural parameters in the predictive objective.

4 Method

4.1 Overview

NSR-KT consists of two tightly coupled modules with a fixed division of labor:

1. a **symbolic module** built from a compression-based predictor that outputs a probabilistic prior over the next response;
2. a **neural residual module** that predicts a correction term for that prior.

The symbolic module is responsible for explicit finite-context regularities, while the neural module is responsible for systematic departures from those regularities. The final predictor is obtained by adding both contributions in logit space. This single-residual formulation is the core unit. The final NSR-KT Convex MoE combines several such residual experts with strictly fold-out-selected convex weights.

4.2 Symbolic module: compression-based prior estimator

For each learner, the symbolic module maintains an online context model over the token sequence $u_{1:t}$. We keep the notation generic and write its predictive distribution as P_{sym} , which may be instantiated by PPM, PST, context mixing, or a related symbolic predictor. Given the upcoming concept k_{t+1} , the symbolic prior p_t^{sym} is computed by Equation (2).

In addition to the prior itself, the symbolic module can expose lightweight context statistics,

$$\mathbf{z}_t = [d_t, \log N(c_t), H(c_t), \rho_t], \quad (7)$$

where d_t is the matched context depth, $N(c_t)$ is the visit count,

$$H(c_t) = - \sum_{a \in \mathcal{A}} P_{\text{sym}}(a | c_t) \log P_{\text{sym}}(a | c_t)$$

is the local count-entropy of the active symbolic context under the backend’s smoothed predictive distribution, and

$$\rho_t = \frac{-\log P_{\text{sym}}(u_t | u_{1:t-1})}{\log |\mathcal{A}|} \quad (8)$$

is the local compression ratio. These quantities are not alternative predictors; they simply summarize how confidently the symbolic model explains the recent trajectory.

4.3 Neural module: residual correction network

The neural branch receives the original educational interaction features and predicts the residual term in Equation (3). Let

$$\mathbf{e}_t = [E_q(q_t); E_k(k_t); E_r(r_t); W_m m_t]$$

be the embedding of the current interaction. A neural backbone such as DKT, SAKT, AKT, or SimpleKT maps $\mathbf{e}_{1:t}$ to a hidden state $\mathbf{h}_t$. We then define the residual network as

$$\delta_t = f_\theta([\mathbf{h}_t; \text{logit}(\tilde{p}_t^{\text{sym}}); \mathbf{z}_t]), \quad (9)$$

where f_θ is a small multilayer perceptron or linear head attached to the neural backbone.

The final prediction is therefore

$$p_t = \sigma(\text{logit}(\tilde{p}_t^{\text{sym}}) + \delta_t). \quad (10)$$

This formulation has a useful inductive bias: if the symbolic prior is already adequate, the optimal residual is close to zero; if the symbolic prior is systematically biased, the neural network learns to correct that bias rather than relearning the full response distribution from scratch.

4.4 Residual expert ensemble

The single-backbone formulation in Equation (10) can be extended without changing the symbolic-prior interface. Suppose we instantiate M residual experts. Expert m produces a symbolic prior $p_{t,m}^{\text{sym}}$, a residual $\delta_t^{(m)}$, and a final prediction

$$p_t^{(m)} = \sigma(\text{logit}(\tilde{p}_{t,m}^{\text{sym}}) + \delta_t^{(m)}). \quad (11)$$

The final NSR-KT Convex MoE combines these residualized predictors using nonnegative global weights selected strictly out of fold:

$$p_t^{\text{ens}} = \sum_{m=1}^M \alpha_m p_t^{(m)}, \quad \alpha_m \geq 0, \quad \sum_{m=1}^M \alpha_m = 1. \quad (12)$$

The weights are chosen on the training folds for each held-out fold, so no held-out labels or item-aware statistics enter the ensemble construction.

For analysis, we also compare global convex aggregation with probability averaging, logit averaging, and a lightweight gated residual committee.

Table 1: Dataset statistics after pyKT preprocessing [Liu et al., 2022]; source datasets are NIPS34 [Wang et al., 2021], Algebra2005 and Bridge2006 from PSLC DataShop/KDD Cup 2010 [Pittsburgh Science of Learning Center DataShop, 2010, Stamper et al., 2010], and ASSIST2009 [Feng et al., 2009, Heffernan and Heffernan, 2014].

Dataset	Interactions	Sequences	Questions	KCs	Avg. length
NIPS34	1,399,470	9,401	948	57	148.86
Algebra2005	884,098	4,712	173,113	112	187.63
Bridge2006	1,817,476	6,043	129,263	493	300.77
ASSIST2009	337,415	4,661	17,737	123	72.39

4.5 Training objective

The residual target δ_t^* is not observed directly, so each NSR-KT residual expert is trained end to end through the final prediction in Equation (10). The objective is

$$\mathcal{L} = - \sum_{t=1}^{T-1} \left[r_{t+1} \log p_t + (1 - r_{t+1}) \log(1 - p_t) \right] + \lambda_\theta \|\theta\|_2^2 + \lambda_{\mathcal{T}} |\mathcal{T}|, \quad (13)$$

where $|\mathcal{T}|$ is the number of active symbolic nodes or contexts. The first term optimizes predictive likelihood, the second regularizes neural complexity, and the third prevents uncontrolled symbolic growth, which acts as a practical surrogate for an MDL-style complexity penalty. When residual aggregation is used, the gating parameters are absorbed into θ and optimized under the same objective.

Importantly, the symbolic and neural modules are optimized differently. The symbolic module is updated online by count-based context maintenance after each observed interaction, while the neural parameters θ are optimized by backpropagation through the residual loss. This asymmetry is intentional: symbolic structure is accumulated explicitly, whereas residual corrections are learned statistically.

5 Experiments

5.1 Experimental setup

We evaluate NSR-KT on four standard public KT benchmarks: NIPS34 from the NeurIPS 2020 Education Challenge [Wang et al., 2021], Algebra2005 and Bridge2006 from KDD Cup 2010/PSLC DataShop [Pittsburgh Science of Learning Center DataShop, 2010, Stamper et al., 2010], and ASSIST2009 from ASSISTments 2009–2010 [Feng et al., 2009, Heffernan and Heffernan, 2014]. The experimental goal is predictive rather than benchmark-diagnostic: we test whether symbolic-prior residualization improves next-response prediction under the pyKT standard test-question-window/late-mean protocol. The experiments focus on residualized DKT, AKT, and SimpleKT prediction networks under PPM, PST, and context-mixing symbolic backends, with a separate ablation for residual-expert aggregation.

We compare residualized DKT, AKT, and SimpleKT backbones above symbolic priors [Piech et al., 2015, Ghosh et al., 2020, Liu et al., 2023]; symbolic-only PPM, PST, and context-mixing predictors without neural residual correction; NSR-KT variants instantiated with PPM, PST, and context-mixing priors; the final NSR-KT Convex MoE with strictly fold-out-selected nonnegative convex weights; and aggregation baselines including probability averaging, logit averaging, and gated residual aggregation.

The external baseline set follows pyKT [Liu et al., 2022] and includes DKT/DKT+ [Piech et al., 2015, Yeung and Yeung, 2018], KQN [Lee and Yeung, 2019], DKVMN [Zhang et al., 2017], ATKT [Guo et al., 2021], GKT [Nakagawa et al., 2019], SAKT [Pandey and Karypis, 2019], SAINT [Choi et al., 2020], AKT [Ghosh et al., 2020], SKVMN [Abdelrahman and Wang, 2019], DeepIRT [Yeung, 2019], SimpleKT [Liu et al., 2023], sparseKT-soft/topK [Huang et al., 2023], RKT [Pandey and Srivastava, 2020], DTransformer [Yin et al., 2023], and recent KT baselines including LPKT, DIMKT, IEKT, QIKT, FoLiBiKT, UKT, DenoiseKT, and csKT [Shen et al., 2021, 2022, Long et al., 2021, Chen et al., 2023, Im et al., 2023, Cheng et al., 2025, Deng et al., 2025, Bai et al., 2025].

We report AUC and accuracy in the main results table to match the external pyKT benchmark protocol. The NSR-KT row reports the best strictly fold-out ensemble variant for each dataset/backend setting.

Table 2: Main predictive AUC results on four KT datasets. All values are reported as test-question-window/late-mean mean $\pm$ std under the pyKT evaluation protocol [Liu et al., 2022]. The NSR-KT row reports mean $\pm$ std over folds for the selected strictly fold-out global-convex ensemble variant; best results are bolded and second-best results are underlined.

Model	NIPS34	Algebra2005	Bridge2006	ASSIST2009
DKT [Piech et al., 2015]	0.7689 $\pm$ 0.0002	0.8149 $\pm$ 0.0011	0.8015 $\pm$ 0.0008	0.7541 $\pm$ 0.0011
DKT+ [Yeung and Yeung, 2018]	0.7696 $\pm$ 0.0002	0.8156 $\pm$ 0.0011	0.8020 $\pm$ 0.0004	0.7547 $\pm$ 0.0017
KQN [Lee and Yeung, 2019]	0.7684 $\pm$ 0.0003	0.8027 $\pm$ 0.0015	0.7936 $\pm$ 0.0014	0.7477 $\pm$ 0.0011
DKVMN [Zhang et al., 2017]	0.7673 $\pm$ 0.0004	0.8054 $\pm$ 0.0011	0.7983 $\pm$ 0.0009	0.7473 $\pm$ 0.0006
ATKT [Guo et al., 2021]	0.7665 $\pm$ 0.0001	0.7995 $\pm$ 0.0023	0.7889 $\pm$ 0.0008	0.7470 $\pm$ 0.0008
GKT [Nakagawa et al., 2019]	0.7689 $\pm$ 0.0024	0.8110 $\pm$ 0.0009	0.8046 $\pm$ 0.0008	0.7424 $\pm$ 0.0021
SAKT [Pandey and Karypis, 2019]	0.7517 $\pm$ 0.0005	0.7880 $\pm$ 0.0063	0.7740 $\pm$ 0.0008	0.7246 $\pm$ 0.0017
SAINT [Choi et al., 2020]	0.7873 $\pm$ 0.0007	0.7775 $\pm$ 0.0017	0.7781 $\pm$ 0.0013	0.6958 $\pm$ 0.0023
AKT [Ghosh et al., 2020]	0.8033 $\pm$ 0.0003	0.8306 $\pm$ 0.0019	0.8208 $\pm$ 0.0007	0.7853 $\pm$ 0.0017
SKVMN [Abdelrahman and Wang, 2019]	0.7513 $\pm$ 0.0005	0.7463 $\pm$ 0.0022	0.7287 $\pm$ 0.0052	0.7332 $\pm$ 0.0009
DeepIRT [Yeung, 2019]	0.7672 $\pm$ 0.0006	0.8040 $\pm$ 0.0013	0.7976 $\pm$ 0.0006	0.7465 $\pm$ 0.0006
SimpleKT [Liu et al., 2023]	0.8035 $\pm$ 0.0000	0.8254 $\pm$ 0.0003	0.8160 $\pm$ 0.0006	0.7744 $\pm$ 0.0018
sparseKT-soft [Huang et al., 2023]	0.7994 $\pm$ 0.0005	0.8156 $\pm$ 0.0028	0.8112 $\pm$ 0.0001	0.7745 $\pm$ 0.0010
sparseKT-topK [Huang et al., 2023]	0.7997 $\pm$ 0.0005	0.8120 $\pm$ 0.0018	0.8094 $\pm$ 0.0005	0.7701 $\pm$ 0.0010
RKT [Pandey and Srivastava, 2020]	0.7966 $\pm$ 0.0011	0.7761 $\pm$ 0.0010	0.7631 $\pm$ 0.0013	0.7628 $\pm$ 0.0070
DTransformer [Yin et al., 2023]	0.7944 $\pm$ 0.0003	0.8188 $\pm$ 0.0025	0.8093 $\pm$ 0.0009	0.7725 $\pm$ 0.0025
LPKT [Shen et al., 2021]	0.8004 $\pm$ 0.0003	0.8268 $\pm$ 0.0004	0.8056 $\pm$ 0.0008	0.7812 $\pm$ 0.0023
DIMKT [Shen et al., 2022]	0.8030 $\pm$ 0.0002	0.8277 $\pm$ 0.0009	0.8167 $\pm$ 0.0008	0.7717 $\pm$ 0.0011
IEKT [Long et al., 2021]	0.8045 $\pm$ 0.0002	<u>0.8416</u> $\pm$ 0.0014	0.8125 $\pm$ 0.0009	0.7861 $\pm$ 0.0027
QIKT [Chen et al., 2023]	0.8044 $\pm$ 0.0005	0.8408 $\pm$ 0.0007	0.8101 $\pm$ 0.0003	0.7878 $\pm$ 0.0024
FoLiBiKT [Im et al., 2023]	0.8033 $\pm$ 0.0002	0.8316 $\pm$ 0.0012	0.8208 $\pm$ 0.0016	0.7838 $\pm$ 0.0013
UKT [Cheng et al., 2025]	0.8027 $\pm$ 0.0004	0.8388 $\pm$ 0.0010	<u>0.8218</u> $\pm$ 0.0019	0.7850 $\pm$ 0.0018
DenoiseKT [Deng et al., 2025]	0.8046 $\pm$ 0.0002	0.8347 $\pm$ 0.0006	0.8129 $\pm$ 0.0005	<u>0.7898</u> $\pm$ 0.0013
csKT [Bai et al., 2025]	<u>0.8055</u> $\pm$ 0.0001	0.8307 $\pm$ 0.0008	0.8211 $\pm$ 0.0009	0.7856 $\pm$ 0.0021
NSR-KT Convex MoE	0.8060 $\pm$ 0.0004	0.8460 $\pm$ 0.0005	0.8269 $\pm$ 0.0010	0.7975 $\pm$ 0.0019

5.2 Implementation details

All baseline and NSR-KT numbers are reported under the pyKT test-question-window/late-mean evaluation protocol. All NSR-KT results use the same pyKT preprocessing, data splits, test-question-window evaluation, and late-mean aggregation protocol. For NSR-KT, each residual expert is trained above the same symbolic-prior interface, and the final ensemble is selected strictly fold-out: for each held-out fold, ensemble weights are selected only on the other folds and then applied to the held-out fold. This protocol prevents held-out labels from entering the ensemble construction.

5.3 Main results

Tables 2 and 3 report the final NSR-KT Convex MoE against ordinary KT baselines. NSR-KT reaches AUC/accuracy $0.805956 \pm 0.000432/0.734990 \pm 0.000615$ on NIPS34/PST, $0.845960 \pm 0.000539/0.825761 \pm 0.000552$ on Algebra2005/ContextMix, $0.826937 \pm 0.000990/0.859383 \pm 0.001068$ on Bridge2006/PST, and $0.797456 \pm 0.001870/0.746699 \pm 0.000878$ on ASSIST2009/ContextMix. It gives the best AUC on all four datasets and the best accuracy on all four datasets among the listed baselines.

Table 4 gives the direct fair ensemble comparison on ASSIST2009: adding the NSR-KT symbolic prior and residualization improves the neural ensemble by 0.0085 AUC and 0.0055 accuracy.

Additional ablations. Detailed symbolic-prior ablations and residual aggregation variants are moved to Appendix C to keep the main narrative focused on the headline comparison and the held-out case study.

5.4 Residual aggregation ablation

We next test whether the residual interface can support multiple correction experts. Across dataset/backend pairs, the fold-out convex MoE improves over the strongest named single residual backbone on both AUC and accuracy. The largest gains appear on Algebra2005; ASSIST2009 also shows consistent improvements, while NIPS34 gains are smaller but remain positive. This indicates

Table 3: Main predictive accuracy results on four KT datasets. All values are reported as test-question-window/late-mean mean $\pm$ std under the pyKT evaluation protocol [Liu et al., 2022]. The NSR-KT row reports mean $\pm$ std over folds for the selected strictly fold-out global-convex ensemble variant; best results are bolded and second-best results are underlined.

Model	NIPS34	Algebra2005	Bridge2006	ASSIST2009
DKT [Piech et al., 2015]	0.7032 $\pm$ 0.0004	0.8097 $\pm$ 0.0005	0.8553 $\pm$ 0.0002	0.7244 $\pm$ 0.0014
DKT+ [Yeung and Yeung, 2018]	0.7039 $\pm$ 0.0004	0.8097 $\pm$ 0.0007	0.8553 $\pm$ 0.0003	0.7248 $\pm$ 0.0009
KQN [Lee and Yeung, 2019]	0.7028 $\pm$ 0.0001	0.8025 $\pm$ 0.0006	0.8532 $\pm$ 0.0006	0.7228 $\pm$ 0.0009
DKVMN [Zhang et al., 2017]	0.7016 $\pm$ 0.0005	0.8027 $\pm$ 0.0007	0.8545 $\pm$ 0.0002	0.7199 $\pm$ 0.0010
ATKT [Guo et al., 2021]	0.7013 $\pm$ 0.0002	0.7998 $\pm$ 0.0019	0.8511 $\pm$ 0.0004	0.7208 $\pm$ 0.0009
GKT [Nakagawa et al., 2019]	0.7014 $\pm$ 0.0028	0.8088 $\pm$ 0.0008	0.8555 $\pm$ 0.0002	0.7153 $\pm$ 0.0032
SAKT [Pandey and Karypis, 2019]	0.6879 $\pm$ 0.0004	0.7954 $\pm$ 0.0020	0.8461 $\pm$ 0.0005	0.7063 $\pm$ 0.0018
SAINT [Choi et al., 2020]	0.7180 $\pm$ 0.0006	0.7791 $\pm$ 0.0016	0.8411 $\pm$ 0.0065	0.6936 $\pm$ 0.0034
AKT [Ghosh et al., 2020]	0.7323 $\pm$ 0.0005	0.8124 $\pm$ 0.0011	0.8587 $\pm$ 0.0005	0.7392 $\pm$ 0.0021
SKVMN [Abdelrahman and Wang, 2019]	0.6885 $\pm$ 0.0005	0.7837 $\pm$ 0.0023	0.8406 $\pm$ 0.0005	0.7156 $\pm$ 0.0012
DeepIRT [Yeung, 2019]	0.7014 $\pm$ 0.0008	0.8037 $\pm$ 0.0009	0.8543 $\pm$ 0.0003	0.7195 $\pm$ 0.0004
SimpleKT [Liu et al., 2023]	0.7328 $\pm$ 0.0001	0.8083 $\pm$ 0.0005	0.8579 $\pm$ 0.0003	0.7320 $\pm$ 0.0012
sparseKT-soft [Huang et al., 2023]	0.7291 $\pm$ 0.0005	0.8020 $\pm$ 0.0014	0.8569 $\pm$ 0.0005	0.7292 $\pm$ 0.0009
sparseKT-topK [Huang et al., 2023]	0.7287 $\pm$ 0.0003	0.8000 $\pm$ 0.0009	0.8563 $\pm$ 0.0003	0.7271 $\pm$ 0.0010
RKT [Pandey and Srivastava, 2020]	0.7264 $\pm$ 0.0010	0.7964 $\pm$ 0.0021	0.8468 $\pm$ 0.0005	0.7289 $\pm$ 0.0062
DTransformer [Yin et al., 2023]	0.7295 $\pm$ 0.0007	0.8043 $\pm$ 0.0021	0.8555 $\pm$ 0.0007	0.7295 $\pm$ 0.0017
LPKT [Shen et al., 2021]	0.7309 $\pm$ 0.0006	0.8154 $\pm$ 0.0008	0.8547 $\pm$ 0.0005	0.7355 $\pm$ 0.0013
DIMKT [Shen et al., 2022]	0.7312 $\pm$ 0.0005	0.8109 $\pm$ 0.0005	0.8579 $\pm$ 0.0001	0.7354 $\pm$ 0.0019
IEKT [Long et al., 2021]	0.7330 $\pm$ 0.0002	<u>0.8236</u> $\pm$ 0.0010	0.8553 $\pm$ 0.0023	0.7375 $\pm$ 0.0042
QIKT [Chen et al., 2023]	0.7333 $\pm$ 0.0005	0.8222 $\pm$ 0.0006	0.8539 $\pm$ 0.0005	0.7381 $\pm$ 0.0014
FoLiBiKT [Im et al., 2023]	0.7323 $\pm$ 0.0001	0.8135 $\pm$ 0.0016	0.8585 $\pm$ 0.0008	0.7393 $\pm$ 0.0010
UKT [Cheng et al., 2025]	0.7310 $\pm$ 0.0008	0.8173 $\pm$ 0.0015	0.8583 $\pm$ 0.0014	0.7400 $\pm$ 0.0016
DenoiseKT [Deng et al., 2025]	0.7330 $\pm$ 0.0007	0.8178 $\pm$ 0.0009	0.8553 $\pm$ 0.0002	<u>0.7427</u> $\pm$ 0.0007
csKT [Bai et al., 2025]	<u>0.7344</u> $\pm$ 0.0005	0.8118 $\pm$ 0.0013	<u>0.8591</u> $\pm$ 0.0004	0.7402 $\pm$ 0.0004
NSR-KT Convex MoE	0.7350 $\pm$ 0.0006	0.8258 $\pm$ 0.0006	0.8594 $\pm$ 0.0011	0.7467 $\pm$ 0.0009

Table 4: ASSIST2009 ensemble comparison with and without the symbolic prior. The neural ensemble baseline is the best fold-out ensemble over standard DKT, AKT, and SimpleKT predictions. Results are five-fold test-question-window/late-mean mean $\pm$ std.

Setting	AUC	Acc.	Δ AUC	Δ Acc.
Without symbolic prior	0.788925 $\pm$ 0.001739	0.741204 $\pm$ 0.000668	–	–
With symbolic prior (NSR-KT)	0.797456 $\pm$ 0.001870	0.746699 $\pm$ 0.000878	+0.008531	+0.005495

that the MoE acts as a controlled residual-combination step rather than a separate item-memory mechanism. Detailed aggregation equations, weight diagnostics, and fold-out variants are reported in Appendix C.

6 Discussion

NSR-KT exposes a simple prediction path: the symbolic prior summarizes finite-context evidence, while the residual logit measures how much the neural backbone revises that prior. The fold-out convex weights further show which residual experts remain useful without held-out labels or item-aware statistics. Figure 2 gives a held-out question-level example where NSR-KT produces fewer threshold errors than ordinary KT baselines.

Held-out question-level case study. Figure 2 compares NSR-KT Convex MoE with ordinary KT baselines on a NIPS34/PST test-window segment from fold 3, student 370. Over these 30 consecutive questions, NSR-KT obtains 20/30 correct predictions, while SimpleKT and AKT obtain 16/30 and DKT obtains 13/30. Blue boxes mark steps 25, 42, and 52, where all three ordinary KT baselines are wrong but NSR-KT is correct.

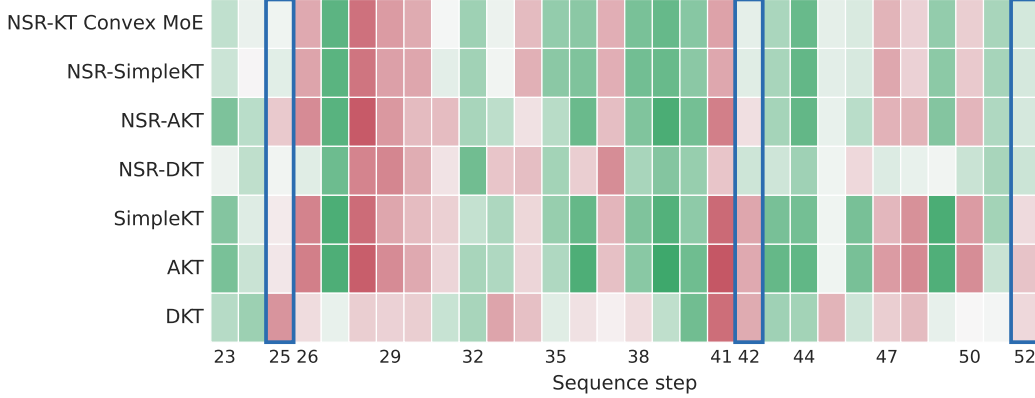


Figure 2: Held-out NIPS34/PST question-level segment (fold 3, student 370, steps 23–52). Each cell is colored by signed confidence relative to the ground-truth label: darker green indicates a more confident correct-side prediction, while darker red indicates a more confident wrong-side prediction. Numeric cell annotations are omitted; blue boxes mark questions where NSR-KT is correct while DKT, AKT, and SimpleKT are all incorrect.

Cold-start protocol and result. We also include the ASSIST2009 prefix cold-start check in the main text because it directly tests the motivation for using a symbolic prior. Training, folds, and fold-out selection are unchanged from the standard protocol; evaluation only keeps held-out prediction targets within each learner’s first 20 interactions, and each target is predicted from the preceding prefix available at that time. Table 5 aggregates each residual backbone over PPM, PST, and ContextMix. NSR-SimpleKT is strongest on average, followed by NSR-AKT and NSR-DKT. At the strongest backend level, ContextMix/NSR-SimpleKT reaches AUC 0.724365 and accuracy 0.708950, compared with 0.720471/0.709347 for ContextMix/NSR-AKT and 0.713594/0.700733 for ContextMix/NSR-DKT. This supports the intended role of the symbolic prior as a stabilizing signal when learner histories are short.

Table 5: ASSIST2009 prefix cold-start single residual summary under the pyKT test-question-window/late-mean protocol.

Residual expert	AUC	Acc.
NSR-SimpleKT	0.720739 $\pm$ 0.004470	0.705795 $\pm$ 0.004610
NSR-AKT	0.718102 $\pm$ 0.002571	0.704062 $\pm$ 0.005674
NSR-DKT	0.710356 $\pm$ 0.004139	0.694757 $\pm$ 0.008838

7 Conclusion

This paper presented NSR-KT, a neuro-symbolic framework that combines compression-based symbolic priors with neural residual correction. Empirically, single-backbone residualization improves matched backbones on the main benchmarks, the fold-out ensemble improves over its residual experts, and the ASSIST2009 cold-start results support the value of the prior-plus-residual design. Overall, these results suggest that separating compressible sequential regularities from neural residual effects is a useful inductive bias for robust knowledge tracing.

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A Appendix A: Proof of Proposition 1 and Residual Mapping

A.1 Notation Table

Table 6: Core notation used by NSR-KT.

Symbol	Meaning
$x_t = (q_t, k_t, r_t, m_t)$	interaction at step t
$\mathbf{x}_{1:T}$	learner trajectory of length T
$u_t = \phi(q_t, k_t, r_t)$	symbolic token at step t
$\mathcal{A}$	symbolic alphabet
$c_t = u_{t-d:t-1}$	active suffix context at step t
D	maximum symbolic context depth
P_{sym}	symbolic predictive distribution over next token
p_t^{sym}	symbolic prior over next correctness
$\tilde{p}_t^{\text{sym}}$	clipped symbolic prior
δ_t	neural residual in logit space
p_t	final NSR-KT correctness probability
$\mathbf{z}_t = [d_t, \log N(c_t), H(c_t), \rho_t]$	symbolic context statistics
θ	neural residual parameters
$\mathcal{T}$	symbolic context structure
$H(\mathcal{X})$	entropy rate of source process $\mathcal{X}$

A.2 Assumption for Asymptotic Information-Theoretic Analysis

Assumption 1 (Stationary ergodic source). *For the asymptotic information-theoretic discussion, the tokenized learner trajectory is modeled as a discrete-time stochastic process $\mathcal{X} = (X_t)_{t \geq 1}$ over a finite alphabet $\mathcal{A}$, and $\mathcal{X}$ is assumed to be stationary and ergodic.*

This assumption is used only to justify the entropy-rate interpretation in the limit $T \rightarrow \infty$. The finite-horizon lower bound below does not require stationarity.

A.3 Proof of Compression-Prediction Equivalence

For a sequential predictor,

$$P_M(X_{1:T}) = \prod_{t=1}^T P_M(X_t \mid X_{1:t-1}).$$

Therefore

$$-\log P_M(X_{1:T}) = -\sum_{t=1}^T \log P_M(X_t \mid X_{1:t-1}). \quad (14)$$

Equation (14) shows that cumulative single-step log-loss is exactly the joint negative log-likelihood of the full sequence. In the Bernoulli KT setting, this is the same decomposition used by the predictive loss in Equation (1).

Under adaptive arithmetic coding, a sequential predictor that assigns probability $P_M(X_t \mid X_{1:t-1})$ to each next symbol induces a code of length $-\log P_M(X_{1:T})$ up to standard implementation constants. Therefore minimizing cumulative cross-entropy or sequence NLL is equivalent to minimizing the description length induced by the predictor, which is the operational content of the compression-prediction equivalence used by NSR-KT.

Proposition 2 (Prediction-compression lower bound). *Let P_T^* be the true distribution of $X_{1:T}$ and let P_M be any sequential model. Then*

$$\frac{1}{T} \mathbb{E}_{P_T^*} [-\log P_M(X_{1:T})] \geq \frac{1}{T} H(X_{1:T}). \quad (15)$$

If Assumption 1 holds, then

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_{P_T^*} [-\log P_M(X_{1:T})] \geq H(\mathcal{X}). \quad (16)$$

Proof. For any horizon T ,

$$\begin{aligned}
\mathbb{E}_{P_T^*}[-\log P_M(X_{1:T})] &= \sum_{x_{1:T}} P_T^*(x_{1:T}) \log \frac{1}{P_M(x_{1:T})} \\
&= \sum_{x_{1:T}} P_T^*(x_{1:T}) \log \frac{P_T^*(x_{1:T})}{P_M(x_{1:T})} - \sum_{x_{1:T}} P_T^*(x_{1:T}) \log P_T^*(x_{1:T}) \\
&= \text{KL}(P_T^* \| P_M) + H(X_{1:T}).
\end{aligned} \tag{17}$$

Since $\text{KL}(P_T^* \| P_M) \geq 0$, Equation (15) follows. Dividing by T and taking $\liminf$ yields Equation (16); under Assumption 1, $\frac{1}{T}H(X_{1:T}) \rightarrow H(\mathcal{X})$. $\square$

Equation (17) is the precise prediction-compression identity used by NSR-KT: expected cumulative log-loss equals source entropy plus a nonnegative mismatch term. Shannon’s source-coding theorem gives the corresponding coding interpretation [Shannon, 1948].

A.4 Residual Mapping and Logit-Space Stability

Proposition 3 (Logit-space residual decomposition). *For any target probability $p_t^* \in (0, 1)$ and any symbolic prior $p_t^{\text{sym}} \in (0, 1)$, there exists a unique residual*

$$\delta_t^* = \text{logit}(p_t^*) - \text{logit}(p_t^{\text{sym}})$$

such that

$$p_t^* = \sigma(\text{logit}(p_t^{\text{sym}}) + \delta_t^*).$$

Moreover, the map

$$F_{p_t^{\text{sym}}}(\delta) = \sigma(\text{logit}(p_t^{\text{sym}}) + \delta)$$

is smooth and strictly increasing on $\mathbb{R}$, with derivative

$$F'_{p_t^{\text{sym}}}(\delta) = F_{p_t^{\text{sym}}}(\delta)(1 - F_{p_t^{\text{sym}}}(\delta)) > 0.$$

Proof. Applying the logit transform to

$$p_t^* = \sigma(\text{logit}(p_t^{\text{sym}}) + \delta_t)$$

gives

$$\text{logit}(p_t^*) = \text{logit}(p_t^{\text{sym}}) + \delta_t,$$

hence

$$\delta_t = \text{logit}(p_t^*) - \text{logit}(p_t^{\text{sym}}).$$

This proves existence and uniqueness. The derivative is strictly positive, so the map is nonsingular and strictly monotone. $\square$

This proposition also clarifies the range of the residual variable used by NSR-KT. The symbolic prior is clipped to $\tilde{p}_t^{\text{sym}} \in [\epsilon, 1 - \epsilon]$, but the residual itself is not constrained to a bounded interval:

$$\delta_t \in \mathbb{R}.$$

The final probability remains in $(0, 1)$ because the sigmoid map sends every real logit to a valid probability.

The logit-space formulation is numerically cleaner than directly adding corrections in probability space. In probability space, an additive correction $p_t = p_t^{\text{sym}} + \eta_t$ must satisfy the data-dependent box constraint

$$-p_t^{\text{sym}} < \eta_t < 1 - p_t^{\text{sym}},$$

which changes with the prior and can become arbitrarily tight near 0 or 1. In contrast, the logit residual lives in the fixed domain $\mathbb{R}$ and preserves a unique inverse map to the final probability.

B Appendix B: Formalization of Symbolic Backends

B.1 Count-to-Probability Conversion and Smoothing

NSR-KT requires each symbolic backend to return a normalized predictive distribution over the next token. For count-based backends, this means turning empirical counts into smoothed probabilities.

For a PST node or root-level fallback, a standard smoothed estimate is the Laplace form

$$\hat{P}(a \mid s) = \frac{N(sa) + \beta}{N(s) + \beta|\mathcal{A}|}, \quad \beta > 0. \quad (18)$$

This guarantees strictly positive probability for every symbol and therefore keeps the induced correctness prior in Equation (2) well defined.

For the binary correctness prior at a fixed next concept, the same idea yields the submission-safe Bernoulli form

$$p_t^{\text{sym}} = \frac{\text{count}(r = 1 \mid c_t, k_{t+1}) + \alpha}{\text{count}(r = 0 \mid c_t, k_{t+1}) + \text{count}(r = 1 \mid c_t, k_{t+1}) + 2\alpha}, \quad \alpha > 0. \quad (19)$$

This is the concrete count-to-probability map behind the symbolic correctness prior when the backend is reduced to binary outcome counts under the queried concept.

B.2 Logit Conversion and Clipping

After smoothing, NSR-KT converts the symbolic prior to logit space through

$$\tilde{p}_t^{\text{sym}} = \text{clip}(p_t^{\text{sym}}, \epsilon, 1 - \epsilon), \quad \text{logit}(\tilde{p}_t^{\text{sym}}) = \log \frac{\tilde{p}_t^{\text{sym}}}{1 - \tilde{p}_t^{\text{sym}}}. \quad (20)$$

The clipping step prevents the undefined cases $\text{logit}(0)$ and $\text{logit}(1)$, and therefore avoids numerical instability when the symbolic model is extremely confident. The residual network does not backpropagate through symbolic counts themselves; rather, it consumes the clipped prior logit as a stable input feature inside the final predictor.

B.3 PPM Details

PPM predicts with the longest matched suffix context and recursively backs off to shorter contexts [Cleary and Witten, 1984]. Let c be the current context and let $S(c) = \{a \in \mathcal{A} : N(ca) > 0\}$. A normalized recursive form is

$$P_{\text{PPM}}(a \mid c) = \begin{cases} \frac{N(ca)}{N(c) + \alpha(c)}, & a \in S(c), \\ \frac{\alpha(c)}{N(c) + \alpha(c)} \frac{P_{\text{PPM}}(a \mid \text{suffix}(c))}{1 - \sum_{b \in S(c)} P_{\text{PPM}}(b \mid \text{suffix}(c))}, & a \notin S(c), \end{cases} \quad (21)$$

where $\alpha(c) > 0$ is the escape mass. Equation (21) makes explicit how counts are turned into a smoothed probability distribution: seen symbols use local counts, while unseen symbols share the escaped probability mass and are evaluated through the shorter suffix. This is the detailed count-to-probability conversion behind the abstract interface used in the main text.

B.4 PST Pruning Criterion

For a PST-style backend, the retained suffix set $\mathcal{S}$ is selected by support and predictive-gain criteria [Ron et al., 1996]. A standard pruning condition keeps a suffix s only if it is sufficiently predictive relative to its parent. One admissible ratio test is

$$\frac{\hat{P}(v \mid s)}{\hat{P}(v \mid \text{parent}(s))} \geq r \quad (22)$$

for some threshold $r > 1$ and some continuation symbol v witnessing nontrivial predictive gain. After pruning, prediction for context c uses the deepest retained suffix

$$s^* = \arg \max_{s \in \mathcal{S} : s \preceq c} |s|, \quad (23)$$

and the predictive distribution is

$$P_{\text{PST}}(a | c) = \hat{P}(a | s^*), \quad \hat{P}(a | s) = \frac{N(sa) + \beta}{N(s) + \beta|\mathcal{A}|}, \quad \beta > 0. \quad (24)$$

B.5 Context Mixing

Context mixing combines several bounded-context experts rather than selecting only one. With expert predictions $P_j(a | c)$ and nonnegative weights $w_{j,t}$, the mixed predictor is

$$P_{\text{CM}}(a | c) = \sum_{j=1}^J \tilde{w}_{j,t} P_j(a | c), \quad \tilde{w}_{j,t} = \frac{w_{j,t}}{\sum_{\ell=1}^J w_{\ell,t}}. \quad (25)$$

This is exactly the backend-agnostic mixture interface assumed by NSR-KT.

B.6 Symbolic Feature Engineering

The symbolic statistics used by NSR-KT are

$$\mathbf{z}_t = [d_t, \log N(c_t), H(c_t), \rho_t], \quad (26)$$

where:

1. d_t is the matched context depth;
2. $N(c_t)$ is the visit count of the active context;
3. $H(c_t) = -\sum_{a \in \mathcal{A}} P_{\text{sym}}(a | c_t) \log P_{\text{sym}}(a | c_t)$ is the empirical local entropy of the active node;
4. $\rho_t = -\log P_{\text{sym}}(u_t | u_{1:t-1}) / \log |\mathcal{A}|$ is the local compression ratio.

These are not separate predictors. They are summary features describing how strongly the symbolic model supports its own prior. This is the exact symbolic feature interface referenced by the residual network in Equation (9).

In implementation form, the residual refinement network receives

$$\delta_t = f_{\theta}([\mathbf{h}_t; \text{logit}(\tilde{p}_t^{\text{sym}}); \mathbf{z}_t]), \quad (27)$$

which makes the role of $\mathbf{z}_t$ explicit: it is concatenated with the neural hidden state and the symbolic prior logit before residual prediction.

B.7 Inference Flow and Online Update Complexity

At inference time, the symbolic side of NSR-KT proceeds in four steps:

1. encode the observed interaction history into symbolic tokens and neural inputs;
2. query the symbolic predictor to obtain $\tilde{p}_t^{\text{sym}}$ and the context statistics $\mathbf{z}_t$;
3. compute the residual δ_t with the neural backbone and form the final probability $p_t = \sigma(\text{logit}(\tilde{p}_t^{\text{sym}}) + \delta_t)$;
4. after observing r_{t+1} , append the new token and update the symbolic context model online.

Proposition 4. *Assume the symbolic context structure has bounded maximum depth D . Then symbolic querying and symbolic updating each require $O(D)$ time per interaction.*

Proof. For a bounded-depth context model, the active suffix chain contains at most $D + 1$ contexts, from the deepest matched suffix to the root. Querying the symbolic predictor visits at most one context per depth level to retrieve counts, escape masses, or expert statistics. Updating after observing a new token modifies only the same bounded suffix chain. Hence the number of symbolic states touched per interaction is linear in D and independent of the total trajectory length. $\square$

This shows that, relative to a standard KT backbone, the symbolic overhead added by NSR-KT is strictly bounded by the context depth.

B.8 Pure PPM Baseline Comparison

Table 7: Pure PPM symbolic baseline.

Dataset	AUC	Acc.
NIPS34	0.6979 ± 0.0002	0.6508 ± 0.0003
Algebra2005	0.7779 ± 0.0009	0.7794 ± 0.0005

Table 7 restores the pure symbolic baseline omitted from the short version. On NIPS34, pure PPM reaches only 0.6979 AUC, far below the residualized systems and the final NSR-KT ensemble. On Algebra2005, pure PPM also remains clearly below the best residualized variants. This confirms that symbolic local-context modeling alone is insufficient and that the neural residual contributes information beyond local counts.

C Appendix C: Neural Architecture and Symbolic Features

C.1 Residual-Space Aggregation

Besides the main convex combination over final probabilities, NSR-KT also admits aggregation directly in residual space. In this variant, experts share the same clipped symbolic prior $\tilde{p}_t^{\text{sym}}$ and differ only through their residual terms:

$$p_t = \sigma\left(\text{logit}(\tilde{p}_t^{\text{sym}}) + A_t\left(\delta_t^{(1)}, \dots, \delta_t^{(M)}\right)\right). \quad (28)$$

A simple convex residual committee is

$$A_t\left(\delta_t^{(1)}, \dots, \delta_t^{(M)}\right) = \sum_{m=1}^M \alpha_m \delta_t^{(m)}, \quad \alpha_m \geq 0, \quad \sum_{m=1}^M \alpha_m = 1. \quad (29)$$

Under the shared-prior formulation, the uniform case coincides with logit averaging of the corresponding final predictors:

$$\text{logit}(\tilde{p}_t^{\text{sym}}) + \frac{1}{M} \sum_{m=1}^M \left[\text{logit}(p_t^{(m)}) - \text{logit}(\tilde{p}_t^{\text{sym}}) \right] = \frac{1}{M} \sum_{m=1}^M \text{logit}(p_t^{(m)}). \quad (30)$$

We therefore treat post-hoc logit averaging as the matched baseline for uniform residual aggregation, and compare it with the fold-out convex MoE used in the main results.

C.2 Lightweight Gating

A lightweight adaptive gate takes the form

$$A_t\left(\delta_t^{(1)}, \dots, \delta_t^{(M)}\right) = \sum_{m=1}^M g_m(\mathbf{s}_t) \delta_t^{(m)}, \quad \sum_{m=1}^M g_m(\mathbf{s}_t) = 1, \quad g_m(\mathbf{s}_t) \geq 0, \quad (31)$$

where the feature vector $\mathbf{s}_t$ includes the prior logit $\text{logit}(\tilde{p}_t^{\text{sym}})$, the symbolic statistics $\mathbf{z}_t$, and disagreement features derived from the residual experts.

C.3 Fold-Out Aggregation Protocol

The ensemble protocol used in the main paper is a static fold-out protocol, not an online test-time gate. For each held-out fold, convex weights are selected using only the remaining folds, and the selected weight vector is then applied unchanged to the held-out fold. Therefore the final ensemble predictor on a test fold is

$$p_t^{\text{ens}} = \sum_{m=1}^M \alpha_m p_t^{(m)}, \quad \alpha_m \geq 0, \quad \sum_{m=1}^M \alpha_m = 1, \quad (32)$$

with the simplex weights α_m fixed for that fold before the held-out labels are evaluated. The average selected weight vectors already reported in Table 9 are the corresponding dataset-level summaries of this protocol.

C.4 Mechanistic Interpretation

NSR-KT is interpretable by construction because the symbolic prior and the residual logit play distinct roles. The symbolic model exposes local evidence through the matched suffix depth, historical counts, local entropy, and compression ratio. The residual term then measures how much the neural backbone revises that symbolic explanation after incorporating denser global information.

This interpretation is consistent with the main-text diagnostics. When the symbolic prior is well supported by deep, frequently visited, low-entropy contexts, the residual is expected to remain comparatively small. When local symbolic evidence is weak or when dense neural features conflict with the local suffix pattern, the residual magnitude increases and the final decision is driven more strongly by the neural correction. The held-out NIPS34/PST case study in Figure 2 illustrates this mechanism: NSR-KT does not replace the neural KT backbones, but uses the symbolic-prior residual interface and fold-out convex combination to correct threshold-level cases where ordinary KT baselines remain on the wrong side of the decision boundary.

In the reported case study, the observed gain is specifically a threshold-level correction pattern: fixed fold-out convex weights move predictions across the 0.5 decision threshold at a small number of questions where DKT, AKT, and SimpleKT are all incorrect. This is the most concrete error pattern currently supported by the available manuscript materials.

C.5 Additional Experimental Results

Symbolic-prior ablation. To isolate the symbolic prior, we compare each ASSIST2009 residual backbone with the same model after removing the symbolic prior; this is separate from the neural ensemble baseline in the main-text ensemble comparison. Table 8 reports the gains; the prior consistently improves AUC by 0.0340–0.0553 and accuracy by 0.0194–0.0272 across all tested backends and residual backbones.

Table 8: ASSIST2009 symbolic prior ablation. Gains are the full residual model minus the same model without the symbolic prior under the test-question-window/late-mean protocol.

Backend	Residual	Δ AUC	Δ Acc.
ContextMix	NSR-SimpleKT	+0.0340	+0.0234
ContextMix	NSR-AKT	+0.0407	+0.0244
ContextMix	NSR-DKT	+0.0489	+0.0224
PPM	NSR-AKT	+0.0510	+0.0254
PPM	NSR-DKT	+0.0535	+0.0261
PPM	NSR-SimpleKT	+0.0553	+0.0272
PST	NSR-SimpleKT	+0.0380	+0.0194
PST	NSR-AKT	+0.0444	+0.0223
PST	NSR-DKT	+0.0473	+0.0235

Table 9: Best fold-out global-convex ensemble variants by dataset/backend. All metrics are reported as mean $\pm$ std over held-out folds under test-question-window/late-mean. Weights are average simplex weights over NSR-SimpleKT, NSR-AKT, and NSR-DKT.

Dataset	Backend	Selection	Scheme	AUC	Acc.	NLL	Weights
NIPS34	PST	Best AUC	global-prob convex	0.805956 $\pm$ 0.000432	0.734990 $\pm$ 0.000615	0.525585 $\pm$ 0.000588	(0.740, 0.170, 0.090)
NIPS34	PST	Best Acc.	global-prob convex	0.805936 $\pm$ 0.000421	0.735029 $\pm$ 0.000580	0.525657 $\pm$ 0.000599	(0.730, 0.145, 0.125)
Algebra2005	ContextMix	Best AUC	global-prob convex	0.845960 $\pm$ 0.000539	0.825761 $\pm$ 0.000552	0.392648 $\pm$ 0.000325	(0.235, 0.315, 0.450)
Algebra2005	ContextMix	Best Acc.	global-logit convex	0.845438 $\pm$ 0.000508	0.825851 $\pm$ 0.000404	0.393800 $\pm$ 0.000384	(0.330, 0.215, 0.455)
Algebra2005	PST	Best AUC	global-prob convex	0.839907 $\pm$ 0.000956	0.822494 $\pm$ 0.001205	0.399284 $\pm$ 0.001338	(0.245, 0.235, 0.520)
Algebra2005	PST	Best Acc.	global-logit convex	0.839570 $\pm$ 0.000554	0.822698 $\pm$ 0.000959	0.400441 $\pm$ 0.000777	(0.330, 0.115, 0.555)
ASSIST2009	ContextMix	Best AUC	global-prob convex	0.797564 $\pm$ 0.001982	0.746554 $\pm$ 0.001127	0.514387 $\pm$ 0.001475	(0.325, 0.295, 0.380)
ASSIST2009	ContextMix	Best Acc.	global-logit convex	0.797456 $\pm$ 0.001870	0.746699 $\pm$ 0.000878	0.514967 $\pm$ 0.001340	(0.360, 0.205, 0.435)
ASSIST2009	PPM	Best AUC/Acc.	global-prob convex	0.797246 $\pm$ 0.001885	0.746485 $\pm$ 0.000858	0.515312 $\pm$ 0.001525	(0.305, 0.300, 0.395)
ASSIST2009	PST	Best AUC	global-prob convex	0.793013 $\pm$ 0.001281	0.744385 $\pm$ 0.001018	0.518307 $\pm$ 0.000800	(0.280, 0.300, 0.420)
ASSIST2009	PST	Best Acc.	global-residual convex	0.792807 $\pm$ 0.001278	0.744607 $\pm$ 0.001057	0.518930 $\pm$ 0.000794	(0.335, 0.195, 0.470)
Bridge2006	ContextMix	Best AUC	global-prob convex	0.821617 $\pm$ 0.008148	0.851736 $\pm$ 0.012943	–	–
Bridge2006	PPM	Best AUC	global-prob convex	0.825342 $\pm$ 0.001173	0.859034 $\pm$ 0.000850	–	–
Bridge2006	PST	Best AUC	global-prob convex	0.826937 $\pm$ 0.000990	0.859383 $\pm$ 0.001068	–	–
Bridge2006	PST	Best Acc.	global-prob convex	0.826529 $\pm$ 0.001179	0.859404 $\pm$ 0.000985	–	–

Expert diversity diagnostic. As a lightweight diagnostic, we also inspect the average weights of an auxiliary gated residual committee over the SimpleKT, AKT, and DKT residual experts. The weights

do not collapse to a single backbone. On Algebra2005/ContextMix, the average SimpleKT/AKT/DKT weights are 0.329/0.334/0.337; on Algebra2005/PST, they are 0.338/0.307/0.356. This supports the interpretation that the residual experts provide complementary corrections rather than merely duplicating one dominant backbone.

D Appendix D: Full MDL Objective Derivation

Let $U = (u_1, \dots, u_T)$ be the symbolic token sequence and let $R = (r_2, \dots, r_{T+1})$ be the correctness sequence. The MDL-style objective from the main text is

$$\mathcal{R}(\mathcal{D}; \theta, \mathcal{T}) = -\log P_{\text{sym}}(U \mid \mathcal{T}) - \log P_{\theta}^{\text{res}}(R \mid U, \mathcal{T}) + L(\mathcal{T}) + L(\theta). \quad (33)$$

The two likelihood terms expand as

$$-\log P_{\text{sym}}(U \mid \mathcal{T}) = -\sum_{t=1}^T \log P_{\text{sym}}(u_t \mid u_{1:t-1}, \mathcal{T}), \quad (34)$$

$$-\log P_{\theta}^{\text{res}}(R \mid U, \mathcal{T}) = -\sum_{t=1}^{T-1} \left[r_{t+1} \log p_t + (1 - r_{t+1}) \log(1 - p_t) \right], \quad (35)$$

where $p_t = \sigma(\text{logit}(\tilde{p}_t^{\text{sym}}) + \delta_t)$.

Minimizing Equation (33) is therefore equivalent to maximizing the total codelength savings achieved by the symbolic predictor plus the residual predictor, subject to the complexity penalties for $\mathcal{T}$ and θ . In the present NSR-KT implementation, $L(\mathcal{T})$ is represented by the structural penalty $\lambda_{\mathcal{T}}|\mathcal{T}|$, where $|\mathcal{T}|$ counts active symbolic contexts or nodes, and $L(\theta)$ is represented by the neural complexity penalty $\lambda_{\theta}\|\theta\|_2^2$.

This also yields a KL-divergence view of residual learning. If the symbolic prior defines a baseline predictive distribution and the residual model shifts that baseline to the final predictor p_t , then training the residual branch by NLL is equivalent to minimizing the KL gap between the induced final predictive distribution and the true response distribution, while restricting the correction to pass through the symbolic-prior-plus-logit-residual interface.

At the code-length level, $L(\mathcal{T})$ pays for the symbolic context structure itself, including the retained nodes and their transition statistics, while $L(\theta)$ pays for the neural residual parameters. The current implementation uses explicit surrogate penalties rather than a separate variational code for the weights, so the submitted objective stays faithful to the available implementation details and avoids introducing unverifiable extra assumptions.

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